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Task # 1

Earth mean radius, R = 6,371,000 m.

Initial heading = North = 0°

leg 1 : start → wp1

$\Delta\phi = 38.418603 - 38.418012$

$= 0.000591°$

$\Delta\lambda = -110.775784 - (-110.777203)$

$= 0.001419°$

$\phi_1 = 38.418012 \times \frac{\pi}{180} = 0.6705208015$ c

$\phi_2 = \cancel{0.6705}\ 0.6705311164$ c

$\Delta\phi = 0.0000103148\,96$ c

$\Delta\lambda = 0.0000247\,66222$ c

applying Haversine formula,

$$a = \sin^2\left(\frac{0.0000103148\,96}{2}\right) + \cos(0.6705208015)\cos(0.6705311164)\cdot \sin\left(\frac{0.0000247\,66222}{2}\right)$$

$= 1.26730109 \times 10^{-10}$

Now,

$$c = 2\tan^{-1}\left(\frac{\sqrt{1.2073010\,9\times10^{-10}}}{\sqrt{1-1.2073010\,9\times10^{-10}}}\right)$$

$$c = 0.0000219754\,51$$

Therefore,

$$d = 6371000 \times 0.0000219754\,51$$

$$= 140.07 \text{ m.}$$

and,

$$y = \sin(0.0000247662\,22)\cos(0.6705311164)$$

$$= 0.0000194041\,3$$

$$x = \cos(0.6705208015)\sin(0.6705311164) - \sin(0.6705208015) \times \cos(0.6705311164)\cos(0.0000247662\,22)$$

$$= 0.0000103150\,5$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = 62.0053^{\circ}$$

Initial heading = 0° (North)

Turn = 62.0053° - 0°

Turn Right 62.01°

so, leg 1 command Turn right 62.01° and drive 140.01 m.

leg2 : WP1 → WP2

$\Delta\phi = 38.420167 - 38.418603 = 0.001564°$

$\Delta\lambda = -110.776319 - (-110.775784) = -0.000535°$

$\phi_1 = 0.6705311164^c$

$\phi_2 = 0.6705584133^c$

$\Delta\phi = 0.0000272696950^c$

$\Delta\lambda = \div 0.0000093337511^c$

$$a = \sin^2\left(\frac{0.0000272696950}{2}\right) + \cos(0.6705311164)\cos(0.6705584133)\cdot \sin^2\left(\frac{-0.0000093337411}{2}\right)$$

$$= 1.99661037 \times 10^{-10}$$

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$$c = 2\tan^{-1}\left(\frac{\sqrt{1.99661037 \times 10^{-10}}}{\sqrt{1-1.99661037 \times 10^{-10}}}\right) = 0.000028260293$$

$$d = 6371000 \times 0.000028260293 = 180.05\,m.$$

Now,

$$y = \sin(-0.000009337511)\cos(0.6705584133)$$

$$= -0.000007315170.$$

$$x = \cos(0.6705311167)\sin(0.6705584133) - \sin(0.6705311167).$$
$$\cos(0.6705584133).\cos(-0.000009337511)$$

$$= 0.000027297697$$

$$\theta_2 = \tan^{-1}\left(\frac{y}{x}\right) = -15.0029^\circ$$

$$\theta = 360^\circ - 15.0029^\circ = 344.9971^\circ$$

rover in currently facing 62.0053°

Turn $= 344.9971^\circ - 62.0053^\circ = 282.9918^\circ$

Shorter direction is a left turn

Scanned with CamScanner

Turn left = 282.9918 - 360°

= 77.01°

so, leg2 commands, turn left 77.01° and Drive 180.05 m.

leg3: WP2 - WP3

$\Delta\phi$ = 38.420289 - 38.420167

= 0.000122° = 0.000002129302 c

$\Delta\lambda$ = -110.776246 - (-110.776319)

= 0.000073° = 0.000001274090 c

ϕ_1 = 0.6705584133 c

ϕ_2 = 0.6705605426 c

applying ~~Haversine~~ Haversine formula,

$$a = \sin^2\left(\frac{0.000002129302}{2}\right) + \cos(0.6705584133)\cos(0.6705605426)\sin^2\left(\frac{0.000001274090}{2}\right)$$

$$= 1.38259062 \times 10^{-12}$$

Scanned with CamScanner

$$c = 2\tan^{-1}\left(\frac{\sqrt{1.38259062 \times 10^{12}}}{\sqrt{1 - 1.38259062 \times 10^{12}}}\right)$$

$$= 0.00000235167^{\circ}$$

Therefore,

$$d = 6371000 \times 0.00000235167 \rightarrow = 14.98\,m$$

Now,

$$y = \sin(0.00001274090)\cos(0.6705605426)$$

$$= 0.00000998216^{\circ}$$

$$x = \cos(0.6705584133)\sin(0.6705605426) - \sin(0.6705584133)\cos(0.6705605426)\cos(0.00001274090)$$

$$x = 0.00000212930^{2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = 25.1172^{\circ}$$

The rover is currently facing 344.9971°

Turn $= 25.1172^{\circ} - 344.9971^{\circ} = -319.8799^{\circ}$

Scanned with CamScanner

The shorter direction is a right turn

Turn = $-319.8799 + 360 = 40.12°$

So,

leg 3 commands Turn right $40.12°$ and drive 14.98 m.